

EC400: Optimization and Fixed Points

Lecture 1: Tools For Optimization

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A Pre-Sessional Course

Introduction

Lecture 1: Tools for Optimization

- Quadratic Forms and Determinants
- Taylor Theorem
- Concavity and Quasi-Concavity

Lecture 2: Optimization

- Unconstrained Optimization
- Equality Constraints and Lagrange Theorem
- Inequality Constraints and Kuhn-Tucker Theorem

Lecture 3: Comparative Statics and Fixed Points

- Envelope Theorem
- Implicit Function Theorem
- Correspondences and Fixed Points
- Theorem of the Maximum

- **Contact Information:**

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- **Lectures:**

- SOFP Lecture 1 – 9:30-12:00 – Monday 31 August
- SOFP Lecture 2 – 9:30-12:00 – Tuesday 1 September
- SOFP Lecture 3 – 9:30-12:00 – Wednesday 2 September
- SOFP Revision OH – 12:30-13:30 – Friday 18 September

OH by appointment for pastoral concerns

- **Course Recordings & Materials:**

- Can be found on **Moodle**
- Need to be studied before the relevant **lecture**

Suggested Textbooks

- Velleman, *How to Prove It*, 2019.
- Strang, *Introduction to Linear Algebra*, 2023.
- Simon & Blume, *Mathematics for Economists*, 1994.
- Takayama, *Mathematical Economics*, 1985.
- Ok, *Real Analysis with Economic Applications*, 2007.
- Chiang, *Elements of Dynamic Optimization*, 1992.
- Acemoglu, *Introduction to Modern Economic Growth*, 2009.

Tools for Optimization:

Quadratic Forms and Determinants

Defining Quadratic Forms

Quadratic Form Definition

Definition

A quadratic form on $\mathbb{R}^n$ is a real valued function

$$Q(x_1, x_2, \dots, x_n) = \sum_{i \leq j} a_{ij} x_i x_j.$$

Quadratic forms are **useful because**:

- (i) The simplest functions with a global maxima/minima;
- (ii) Conditions for optimization are stated in terms of quadratic forms;
- (iii) Economic optimization problems have a quadratic objective function, such as risk minimization problems in finance, where riskiness is measured by the variance of the returns on investments.

Level Curves of Quadratic Forms

The **level curve** in $\mathbb{R}^2$ can take the form of an ellipse, a hyperbola, a pair of lines, a point, or possibly, the empty set and is

$$a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2 = b.$$

A general quadratic form in $\mathbb{R}^2$

$$a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2,$$

can be written (non uniquely) as

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Matrix Representation of a Quadratic Form

Each quadratic form can be represented as

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

where A is a **unique symmetric** matrix

$$\begin{pmatrix} a_{11} & a_{12}/2 & \dots & a_{1n}/2 \\ a_{12}/2 & a_{22} & \dots & a_{2n}/2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n}/2 & a_{2n}/2 & \dots & a_{nn} \end{pmatrix}.$$

Conversely, the real valued function $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is a quadratic form for any matrix A .

Definiteness of a Quadratic Form

Definiteness of a Quadratic Form

Signing of a Quadratic Form:

A quadratic form always takes the value 0 when $\mathbf{x} = \mathbf{0}$.

We want to know whether $\mathbf{x} = \mathbf{0}$ is a max, a min, or neither.

A Simple Example:

Consider the quadratic form

$$y = ax^2.$$

If $a > 0$, ax^2 is non-negative and equals 0 only when $x = 0$.

If so, the quadratic form is **positive definite** and 0 is a global min.

If $a < 0$, ax^2 is non-positive and equals 0 only when $x = 0$.

If so, the quadratic form is **negative definite** and 0 is a global max.

Definiteness and Semidefiniteness of Symmetric Matrices

Consider **symmetric matrix** A and its associated quadratic form:

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

Definition

Let A be an $(n \times n)$ symmetric matrix. Then A is:

- **positive definite** if $\mathbf{x}^T A \mathbf{x} > \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$
- **positive semidefinite** if $\mathbf{x}^T A \mathbf{x} \geq \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$
- **negative definite** if $\mathbf{x}^T A \mathbf{x} < \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$
- **negative semidefinite** if $\mathbf{x}^T A \mathbf{x} \leq \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$
- **indefinite** if $\mathbf{x}^T A \mathbf{x} > \mathbf{0}$ and $\mathbf{y}^T A \mathbf{y} < \mathbf{0}$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$

Examples in Two Dimensions

In two dimensions, for instance:

- a **positive definite** quadratic form is: $(x_1^2 + x_2^2)$
since it is always positive for $\mathbf{x} \neq \mathbf{0}$.
- a **negative definite** quadratic form is: $-(x_1^2 + x_2^2)$
since it is always negative for $\mathbf{x} \neq \mathbf{0}$.
- a **positive semidefinite** quadratic form is: $(x_1 + x_2)^2$
since it is always non-negative, but equals zero for $\mathbf{x} \neq \mathbf{0}$.
- a **negative semidefinite** quadratic form is: $-(x_1 + x_2)^2$
since it is always non-positive, but equals zero for $\mathbf{x} \neq \mathbf{0}$.
- an **indefinite** quadratic form is: $(x_1^2 - x_2^2)$
since it can take both positive and negative values.

The Determinant

The Determinant

Consider a real-valued matrix A , and denote its i^{th} column by A_i so that

$$A = (A_1, A_2, \dots, A_n).$$

Definition

The **determinant of A** is the unique scalar:

- 1 equal to 1 for the identity matrix,

$$\det(I) = 1;$$

- 2 satisfying linearity in each column,

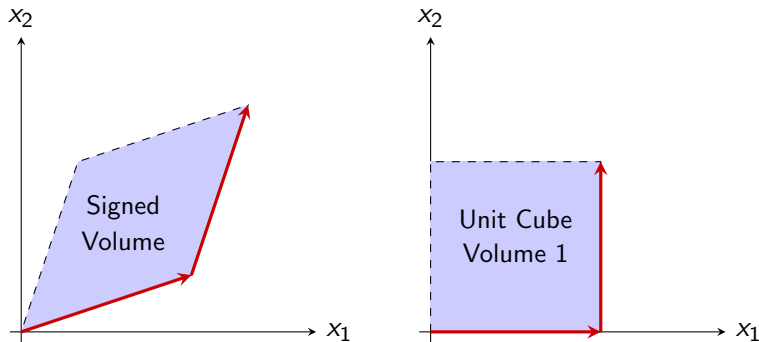
$$\det(aA_1 + B_1, A_2, \dots, A_n) = a \det(A_1, A_2, \dots, A_n) + \det(B_1, A_2, \dots, A_n).$$

- 3 that changes the sign when switching two columns,

$$\det A = -\det(A_2, A_1, A_3, \dots, A_n).$$

The Determinant as a Signed Volume

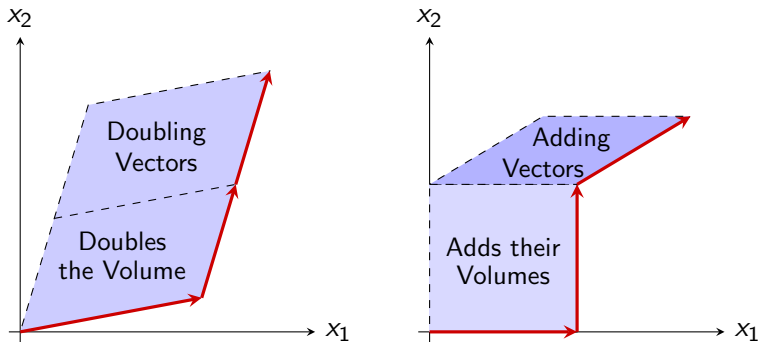
The determinant simply amounts to a signed measure of volume:



$$\det(I) = 1.$$

The Determinant as a Signed Volume

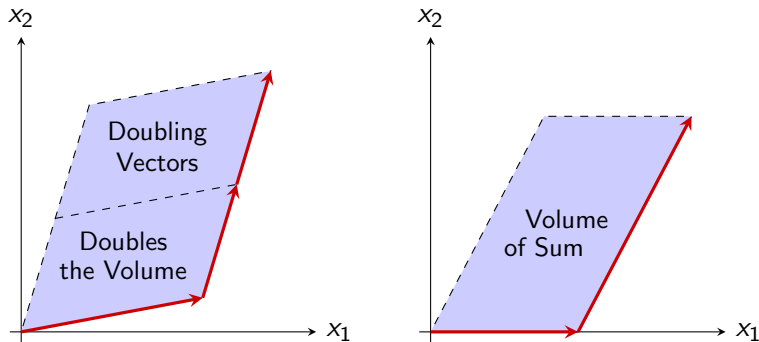
The determinant simply amounts to a signed measure of volume:



$$\det(A_1, A_2 + B_2) = \det(A_1, A_2) + \det(A_1, B_2).$$

The Determinant as a Signed Volume

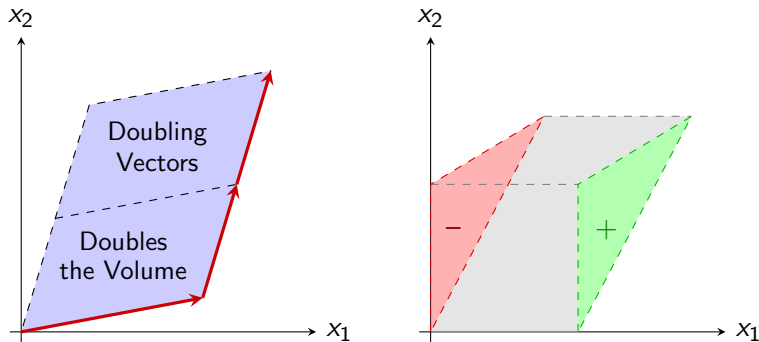
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The Determinant as a Signed Volume

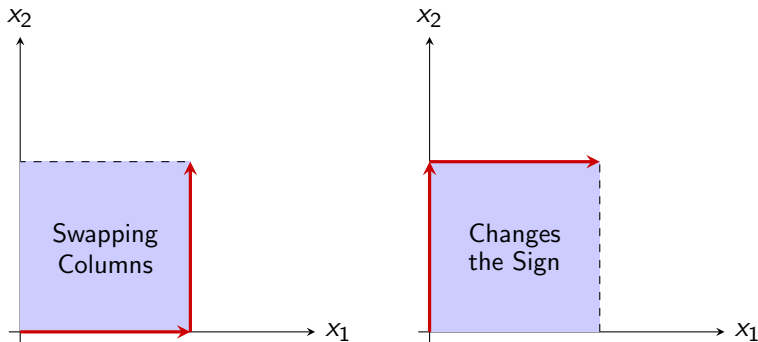
The determinant simply amounts to a signed measure of volume:



$$\det(A_1, A_2 + B_2) = \det(A_1, A_2) + \det(A_1, B_2).$$

The Determinant as a Signed Volume

The determinant simply amounts to a signed measure of volume:



$$\det A = -\det(A_2, A_1).$$

Computing Determinants: Examples

The determinant of a (2×2) matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

is computed as follows $\det(A) = |A| = a_{11}a_{22} - a_{12}a_{21}$.

The determinant of a (3×3) matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

is computed as follows

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Computing Determinants: Laplace Formula

To compute determinants recursively, let:

- a_{ij} be the entry in row i and column j of a matrix A ;
- M_{ij} be the submatrix obtained by deleting row i and column j from A .

Theorem

Laplace formula states that

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij})$$

Principal Submatrices and Principal Minors

Definition

Let A be a $n \times n$ matrix:

- The $k \times k$ submatrix of A formed by deleting $n - k$ columns and the same $n - k$ rows is called a k^{th} order **principal submatrix**.
- The determinant of a k^{th} order principal submatrix is called a k^{th} order **principal minor**.
- The principal submatrix of A obtained by deleting the *last* $n - k$ rows and columns is called the k^{th} order **leading principal submatrix**.
- The determinant of the k^{th} order leading principal submatrix is called the k^{th} order **leading principal minor**.

Example Principal Minors

Consider a general (3×3) matrix A :

- There is **one third order principal minor**: $|A|$.
- There are **three second order principal minors**:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, \quad \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}.$$

The first of these is the leading principal minor.

- There are **three first order principal minors**:

$$a_{11}, \quad a_{22}, \quad a_{33}.$$

The first of these is the leading principal minor.

Testing for Definiteness

Theorem

Let A be an $n \times n$ symmetric matrix. If so, A is:

- positive definite if and only if *all leading principal minors* are positive;
- negative definite if and only if *all leading principal minors* of odd order are negative and all leading principal minors of even order are positive;
- positive semidefinite if and only if *all principal minors* are non-negative;
- negative semidefinite if and only if *all principal minors* of odd order are non-positive and all principal minors of even order are non-negative.

Intuition: Diagonal Matrices

Consider the following 3×3 **diagonal** matrix A

$$A = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix}$$

which corresponds to the simplest quadratic forms

$$a_1 x_1^2 + a_2 x_2^2 + a_3 x_3^2.$$

Such quadratics forms are:

- **positive definite** if and only if all the a_i are **positive**.
- **negative definite** if and only if all the a_i are **negative**.
- **positive semidefinite** if and only if all the a_i are **non-negative**.
- **negative semidefinite** if and only if all the a_i are **non-positive**.
- **indefinite** if there are **two a_i of opposite sign**.

Proof: 2×2 Matrices

Consider the 2×2 symmetric matrix:

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

The associated quadratic form is:

$$\begin{aligned} Q(x_1, x_2) &= \mathbf{x}^T A \mathbf{x} = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= ax_1^2 + 2bx_1x_2 + cx_2^2. \end{aligned}$$

If $a \neq 0$, add and subtract $b^2x_2^2/a$ to get:

$$\begin{aligned} Q(x_1, x_2) &= a \left(x_1^2 + \frac{2bx_1x_2}{a} + \frac{b^2}{a^2}x_2^2 \right) - \frac{b^2}{a}x_2^2 + cx_2^2 \\ &= a \left(x_1 + \frac{b}{a}x_2 \right)^2 + \left(\frac{ac - b^2}{a} \right) x_2^2. \end{aligned}$$

Proof: 2×2 Matrices

Recall that for $a \neq 0$:

$$Q(x_1, x_2) = a \left(x_1 + \frac{b}{a} x_2 \right)^2 + \left(\frac{ac - b^2}{a} \right) x_2^2.$$

If $a = 0$, then Q **cannot be definite** since $Q(1, 0) = 0$.

If coefficients $a > 0$ and $(ac - b^2)/a > 0$ are positive, then Q :

- cannot be negative;
- equals 0 only when $x_1 + b x_2 / a = 0$ and $x_2 = 0$ (ie $x_1 = x_2 = 0$);
- and is thus **positive definite**.

Moreover, Q is positive definite only if both:

$$a > 0 \quad \text{and} \quad \det(A) = ac - b^2 > 0.$$

Proof: 2×2 Matrices

Similarly, if coefficients $a < 0$ and $(ac - b^2)/a < 0$ are negative, then Q :

- cannot be positive;
- equals 0 only when $x_1 + b x_2 / a = 0$ and $x_2 = 0$;
- and is thus **negative definite**.

Moreover, Q is negative definite only if both:

$$a < 0 \quad \text{and} \quad \det(A) = ac - b^2 > 0.$$

If $(ac - b^2) < 0$ instead:

- the two coefficients have **opposite signs**;
- and Q is **indefinite**.

Examples: Testing Definiteness

Consider the quadratic form: $Q(\mathbf{x}) = bx_1^2 + a(x_2^2 + x_3^2) + 2bx_2x_3$.

The corresponding matrix of coefficients amounts to:

$$A = \begin{pmatrix} b & 0 & 0 \\ 0 & a & b \\ 0 & b & a \end{pmatrix}$$

$Q(\mathbf{x})$ is **positive semidefinite** if all principal minors are non-negative:

$$a \geq 0, \quad b \geq 0, \quad ab \geq 0, \quad a^2 - b^2 \geq 0, \quad a^2b - b^3 \geq 0.$$

$Q(\mathbf{x})$ is **negative semidefinite** if all principal minors alternate:

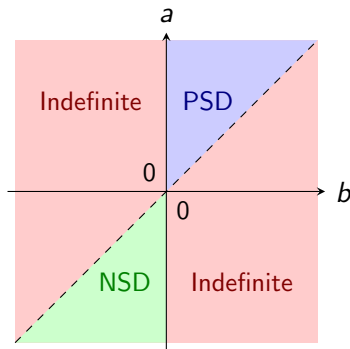
$$a \leq 0, \quad b \leq 0, \quad ab \geq 0, \quad a^2 - b^2 \geq 0, \quad a^2b - b^3 \leq 0.$$

Examples: Testing Definiteness

Therefore $Q(\mathbf{x})$ is:

- **positive semidefinite** if and only if $a \geq 0$, $b \geq 0$, $a \geq b$;
- **negative semidefinite** if and only if $a \leq 0$, $b \leq 0$, $b \geq a$;
- **indefinite** if and only if $ab < b^2$.

Graphically we have that:



Tools for Optimization:

Taylor Approximation

Taylor Expansion

First Order Taylor Expansion

Definition

The first order **Taylor Expansion** of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at a is

$$P(h|a) = f(a) + f'(a)h.$$

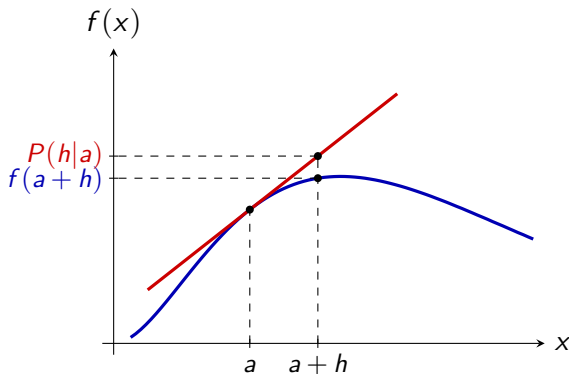
Define the **residual** from the approximation as:

$$R(h|a) = f(a+h) - f(a) - f'(a)h.$$

The approximation holds as the definition of the derivative $f'(a)$ implies

$$\lim_{h \rightarrow 0} \frac{R(h|a)}{h} = 0.$$

First Order Taylor Expansion



Geometrically: TE approximates $f(x)$ by its tangent line at $(a, f(a))$.

Analytically: TE is the best approximation of f by polynomial of degree 1.

Higher Order Taylor Expansions

Definition

The k^{th} order **Taylor Expansion** of $f(x)$ at $x = a$ is defined by

$$P_k(h|a) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \dots + \frac{f^{[k]}(a)}{k!}h^k.$$

The corresponding k^{th} order residual is defined by

$$R_k(h|a) = f(a+h) - P_k(h|a).$$

Taylor's Theorem establishes that

$$\lim_{h \rightarrow 0} \frac{R_k(h|a)}{h^k} = 0.$$

Taylor Expansion: Examples

Consider the exponential function $f(x) = e^x$ at $x = 0$.

All the derivatives of $f(x)$ at $x = 0$ equal 1.

The first and second order Taylor expansions of $f(x)$ are:

- $P_1(h|0) = 1 + h$;
- $P_2(h|0) = 1 + h + \frac{h^2}{2}$.

For $h = 0.2$, then we have that:

- $P_1(0.2|0) = 1.2$;
- $P_2(0.2|0) = 1.22$;
- $f(0.2) = 1.22140275816017$.

Multivariate Taylor Expansion

Definition

The first order **Taylor Expansion** of $F : \mathbb{R}^n \rightarrow \mathbb{R}$ is

$$P_1(\mathbf{h}|\mathbf{a}) = F(\mathbf{a}) + \frac{\partial F}{\partial x_1}(\mathbf{a})h_1 + \dots + \frac{\partial F}{\partial x_n}(\mathbf{a})h_n = F(\mathbf{a}) + DF(\mathbf{a})\mathbf{h},$$

where $DF(\mathbf{a}) = (\partial F / \partial x_1, \dots, \partial F / \partial x_n)$ is the **Jacobian** of F at $\mathbf{a}$.

Alternatively, let $R_1(\mathbf{h}|\mathbf{a})$ denote the residual, to get

$$F(\mathbf{a} + \mathbf{h}) = F(\mathbf{a}) + DF(\mathbf{a})\mathbf{h} + R_1(\mathbf{h}|\mathbf{a}).$$

As before, by Taylor's Theorem, $R_1(\mathbf{h}|\mathbf{a})/||\mathbf{h}|| \rightarrow 0$ as $\mathbf{h} \rightarrow \mathbf{0}$.

Multivariate Taylor Expansion

Definition

The second order **Taylor Expansion** of $F : \mathbb{R}^n \rightarrow \mathbb{R}$ is

$$P_2(\mathbf{h}|\mathbf{a}) = F(\mathbf{a}) + DF(\mathbf{a})\mathbf{h} + \frac{1}{2!}\mathbf{h}^T D^2F(\mathbf{a})\mathbf{h},$$

where $D^2F(\mathbf{a})$ is the **Hessian matrix** of F at $\mathbf{a}$:

$$D^2F(\mathbf{a}) = \begin{pmatrix} \frac{\partial^2 F}{\partial x_1 \partial x_1}(\mathbf{a}) & \cdots & \frac{\partial^2 F}{\partial x_n \partial x_1}(\mathbf{a}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 F}{\partial x_1 \partial x_n}(\mathbf{a}) & \cdots & \frac{\partial^2 F}{\partial x_n \partial x_n}(\mathbf{a}) \end{pmatrix}.$$

The extension for **order k** then trivially follows.

Tools for Optimization:

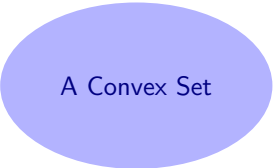
Concavity and Quasi-Concavity

Concavity and Convexity

Definition

A set U is **convex** if for all $\mathbf{x}, \mathbf{y} \in U$ and for all $t \in [0, 1]$

$$t\mathbf{x} + (1 - t)\mathbf{y} \in U.$$



A Convex Set



Not A Convex Set

Concave and Convex Functions

Definition

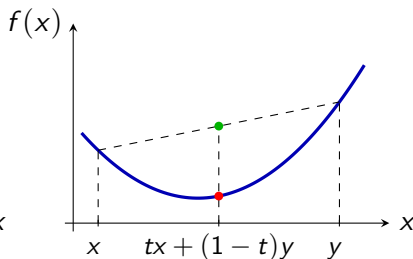
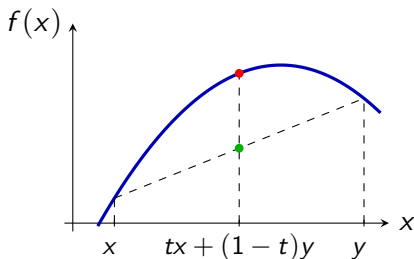
A real valued function f defined on a convex subset U of $\mathbb{R}^n$ is:

(1) **concave** if for all $\mathbf{x}, \mathbf{y} \in U$ and for all $t \in [0, 1]$:

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq tf(\mathbf{x}) + (1-t)f(\mathbf{y});$$

(2) **convex** if for all $\mathbf{x}, \mathbf{y} \in U$ and for all $t \in [0, 1]$:

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y}).$$



Properties of Concavity

Remark

The following trivially hold:

- *The function f is concave if and only if $-f$ is convex.*
- *Linear functions are both convex and concave.*
- *Concave and convex functions must to have convex sets as domains.*

Remark

Let f be a differentiable function on a convex subset U of $\mathbb{R}^n$.

*If so, f is **concave** on U **if and only if** for all $\mathbf{x}, \mathbf{y} \in U$*

$$f(\mathbf{y}) - f(\mathbf{x}) \leq Df(\mathbf{x})(\mathbf{y} - \mathbf{x}).$$

Properties of Concavity

Proof of Remark (If): The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is concave if and only if

$$\begin{aligned} t f(y) + (1 - t) f(x) &\leq f(ty + (1 - t)x) \Leftrightarrow \\ f(x) + t(f(y) - f(x)) &\leq f(x + t(y - x)) \Leftrightarrow \\ f(y) - f(x) &\leq \frac{f(x + t(y - x)) - f(x)}{t}. \end{aligned}$$

For $h = t(y - x)$, the latter is equivalent to

$$f(y) - f(x) \leq \frac{f(x + h) - f(x)}{h}(y - x).$$

The limit of the last expression when $h \rightarrow 0$ yields

$$f(y) - f(x) \leq f'(x)(y - x).$$

□

Properties of Concavity

Proof of Remark (Only If): Let $z = ty + (1 - t)x$.

Recall that concavity implies, for all $x, y \in \mathbb{R}$:

$$f(y) - f(x) \leq f'(x)(y - x).$$

Applying this at z gives:

$$f(y) - f(z) \leq f'(z)(y - z),$$

$$f(x) - f(z) \leq f'(z)(x - z).$$

Multiply the first by t , the second by $(1 - t)$, and sum:

$$t[f(y) - f(z)] + (1 - t)[f(x) - f(z)] \leq 0.$$

This is equivalent to $tf(y) + (1 - t)f(x) \leq f(z)$. $\square$

Properties of Concavity

Remark

A twice differentiable function f on an convex subset U of $\mathbb{R}^n$ is **concave if and only if $D^2f(\mathbf{x})$ is negative semidefinite** for all $\mathbf{x} \in U$.

Proof of Remark (If): The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is concave if and only if

$$f(y) - f(x) \leq f'(x)(y - x) \quad \text{and} \quad f(x) - f(y) \leq f'(y)(x - y).$$

Summing these and dividing by $-(y - x)^2$ implies that

$$\frac{f'(y) - f'(x)}{y - x} \leq 0.$$

The limit of the last expression when $y \rightarrow x$ yields

$$f''(x) \leq 0.$$



Proof of Remark (Only If): Pick $y \geq x$.

By the Fundamental Theorem of Calculus and $f'' \leq 0$:

$$f(y) - f(x) = \int_x^y f'(t) dt.$$

Since $f'' \leq 0$:

$$\int_x^y f'(t) dt \leq \int_x^y f'(x) dt = f'(x)(y - x).$$

Thus $f(y) - f(x) \leq f'(x)(y - x)$, and f is concave. $\square$

Properties of Concavity

Remark

Let $a_1, a_2, \dots, a_k$ be positive number, and let $f_1, \dots, f_k$ be concave functions, each defined on the same convex subset U of $\mathbb{R}^n$.

If so, the function $a_1 f_1 + a_2 f_2 + \dots + a_k f_k$ is **concave** on U .

Proof of Remark: Consider $\mathbf{x} \in U$, $\mathbf{y} \in U$, and $t \in [0, 1]$.

For notational ease, let $k = 2$. If so,

$$\begin{aligned} & a_1 f_1(t\mathbf{x} + (1-t)\mathbf{y}) + a_2 f_2(t\mathbf{x} + (1-t)\mathbf{y}) && \geq \\ \geq & a_1 [t f_1(\mathbf{x}) + (1-t) f_1(\mathbf{y})] + a_2 [t f_2(\mathbf{x}) + (1-t) f_2(\mathbf{y})] && = \\ = & t [a_1 f_1(\mathbf{x}) + a_2 f_2(\mathbf{x})] + (1-t) [a_1 f_1(\mathbf{y}) + a_2 f_2(\mathbf{y})]. \end{aligned}$$

□

Example: Profits

Consider a firm's **profit function**:

- Denote by y output and by $\mathbf{x}$ the input bundle.
- Denote by p the price of output and by w_i the unit-cost of input i .
- The firm's production function is $y = g(\mathbf{x})$.
- The firm's **profit function** amounts to

$$\Pi(\mathbf{x}) = p g(\mathbf{x}) - (w_1 x_1 + w_2 x_2 + \dots + w_n x_n)$$

- **Profits are concave when the production function is concave.**
- This follows by concavity of $g(\mathbf{x})$ and of expenditure function,

$$-(w_1 x_1 + w_2 x_2 + \dots + w_n x_n).$$

Quasi-Concavity and Quasi-Convexity

Definition

Consider a real valued function f defined on $U \subset \mathbb{R}^n$:

(1) its **a-level set** is

$$C_a = \{\mathbf{x} \in U \mid f(\mathbf{x}) = a\};$$

(2) its **a-uppercontour set** is

$$C_a^+ = \{\mathbf{x} \in U \mid f(\mathbf{x}) \geq a\};$$

(3) its **a-lowercontour set** is

$$C_a^- = \{\mathbf{x} \in U \mid f(\mathbf{x}) \leq a\}.$$

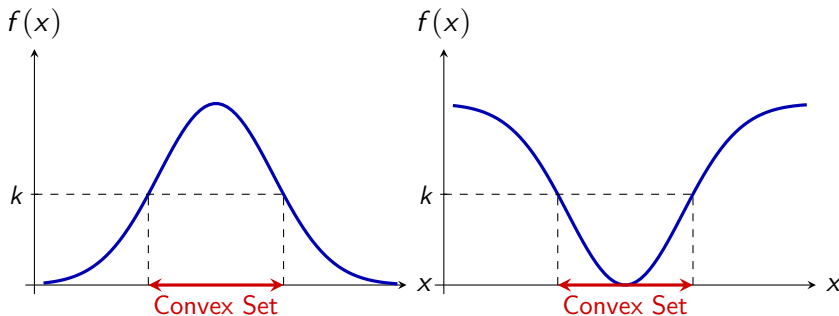
Level sets can be a point, a curve, a plane or an hyperplane.

Quasi-Concavity and Quasi-Convexity

Definition

A real valued function f defined on a convex set $U \subset \mathbb{R}^n$ is:

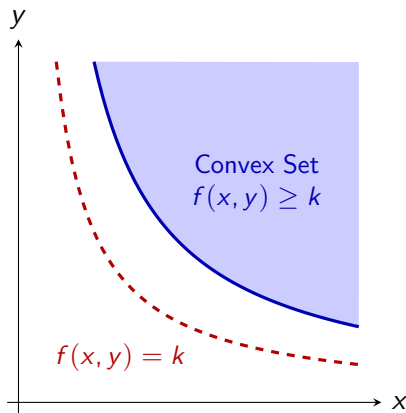
- (1) **quasi-concave** if C_a^+ is a convex set for every real number a ;
- (2) **quasi-convex** if C_a^- is a convex set for every real number a .



Quasi-Concavity in $\mathbb{R}^2$

Level sets of quasi-concave functions bound convex subsets from below.

Level sets of quasi-convex functions bound convex subsets from above.



Properties of Quasi-Concavity

Remark

Every *concave* function is *quasi-concave*.

Every *convex* function is *quasi-convex*.

Proof of Remark: Let f be a concave function and take any two points $\mathbf{x}$ and $\mathbf{y}$ in C_a^+ .

If so, $f(\mathbf{x}) \geq a$ and $f(\mathbf{y}) \geq a$ and so:

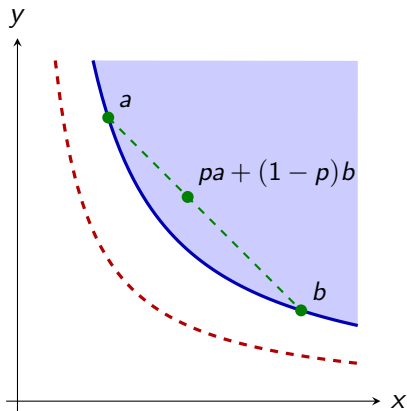
$$\begin{aligned} f(t\mathbf{x} + (1-t)\mathbf{y}) &\geq tf(\mathbf{x}) + (1-t)f(\mathbf{y}) \\ &\geq ta + (1-t)a = a. \end{aligned}$$

Thus, the set C_a^+ is convex since $t\mathbf{x} + (1-t)\mathbf{y} \in C_a^+$. $\square$

Economic Properties: Preferring Mixtures

Quasi-concavity requires upper contour sets to be convex and such a property is often deemed natural in many economic environments.

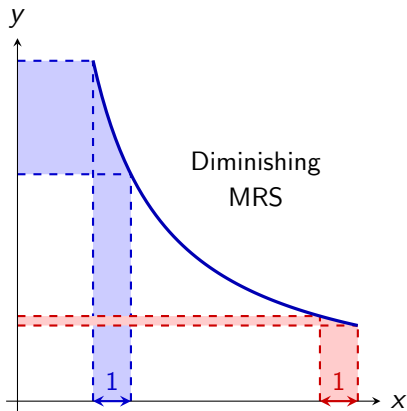
If preferences are quasi-concave, given any two bundles, **a consumer prefers a mixture between the bundles to any one of them.**



Economic Properties: Diminishing MRS

A more fundamental advantage of concavity is that indifference curves display a **diminishing marginal rate of substitution**.

As a consumer increases his consumption for one of the goods, his willingness to pay for an additional unit decreases.

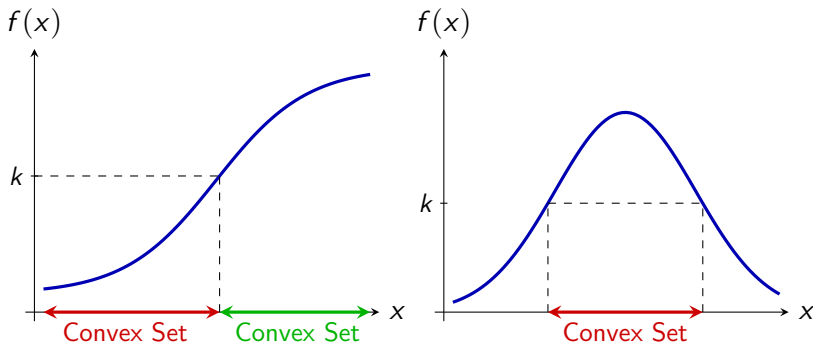


Properties of Quasi-Concavity in $\mathbb{R}$

Remark

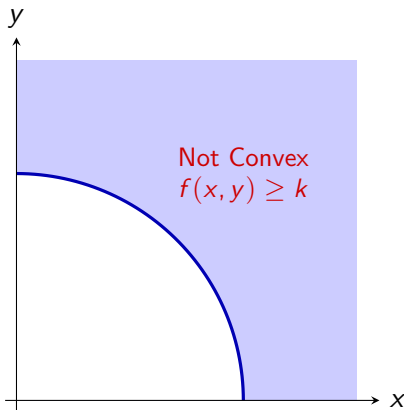
Every *monotone function* on $\mathbb{R}$ is *both quasi-concave and quasi-convex*.

Every *single-peaked function* on $\mathbb{R}$ is *quasi-concave*.



Properties of Quasi-Concavity in $\mathbb{R}^n$

In $\mathbb{R}^n$, **not every monotone function is quasi-concave**.



Example: $f(x, y) = x^2 + y^2$ on $\mathbb{R}^2_+$.

Properties of Quasi-Concavity in $\mathbb{R}^n$

To see this in $\mathbb{R}^2$, consider the function:

$$f(x, y) = x^2 + y^2.$$

Such function is clearly increasing in $\mathbb{R}_+^2$ as:

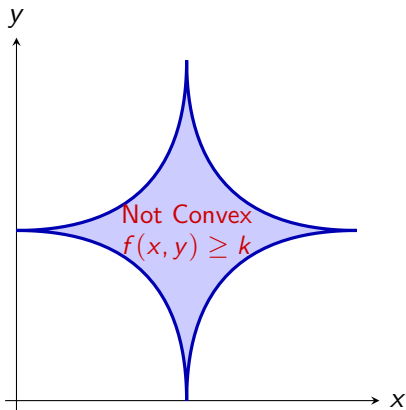
$$Df(x, y) = (2x, 2y).$$

However, the function is not quasi-concave as:

- $(1, 0), (0, 1) \in C_1^+$ as $f(1, 0) = f(0, 1) = 1$;
- $(1/2, 1/2) \notin C_1^+$ as $f(1/2, 1/2) < 1$.

Properties of Quasi-Concavity in $\mathbb{R}^n$

In $\mathbb{R}^n$, not every single-peaked function is quasi-concave.



Example: $f(x, y) = -\sqrt{|x - k|} - \sqrt{|y - k|}$ on $\mathbb{R}^2$.

Remark

Increasing transformations of quasi-concave functions are quasi-concave.

Proof of Remark: Let f be quasi-concave and defined on a convex set U in $\mathbb{R}^n$.

Consider an increasing transformation $g : \mathbb{R} \rightarrow \mathbb{R}$.

For any function h , let $C_k^+(h)$ denote its upper-contour sets.

So if $h(\mathbf{x}) = g(f(\mathbf{x}))$, we have that

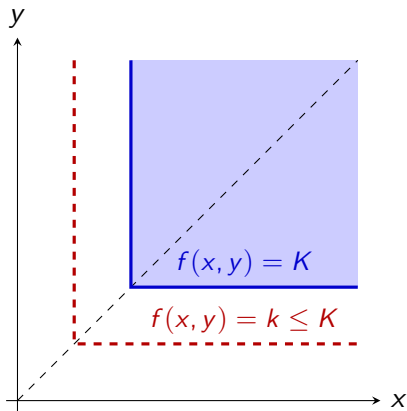
$$C_k^+(f) = C_{g(k)}^+(h).$$

But as $C_k^+(f)$ is convex for all k so is $C_{g(k)}^+(h)$, and h is quasi-concave. $\square$

Examples of Quasi-Concavity

Remark

The Leontief function, $\min\{x, y\}$, is quasi-concave since C_a^+ is convex.



Equivalent Definitions of Quasi-Concavity

Theorem

Let f be a function defined on a convex set U in $\mathbb{R}^n$.

If so, the following statements are equivalent:

- (1) f is a quasi-concave function on U ;
- (2) for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$:

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

Proof of Theorem: To prove (1) implies (2), take any $\mathbf{x}, \mathbf{y} \in U$ and let $\min\{f(\mathbf{x}), f(\mathbf{y})\} = k$.

If so, $\mathbf{x}, \mathbf{y} \in C_k^+$. But as f is quasi-concave, C_k^+ must be convex.

Thus, $t\mathbf{x} + (1 - t)\mathbf{y} \in C_k^+$ for all $t \in [0, 1]$ and

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \geq k = \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

□

Equivalent Definitions of Quasi-Concavity

Proof of Theorem: To prove (2) implies (1), take any $\mathbf{x}, \mathbf{y} \in C_k^+$.

Note that (2) implies that for all $t \in [0, 1]$

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\} \geq k.$$

Thus, $t\mathbf{x} + (1 - t)\mathbf{y} \in C_k^+$ for all $t \in [0, 1]$ and all $\mathbf{x}, \mathbf{y} \in C_k^+$.

But this implies that C_k^+ is convex for all k , and hence f is quasi-concave.

□